

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250281847

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

ZHUANG; Zhi

SAND PAINTING MACHINE

Abstract

A sand painting machine is provided, including a fixed chassis provided with a first rotating driving device, a rotating disc provided with a second rotating driving device, a rotating arm having a second end provided with a magnet and a sand board placed with sand and balls, which are arranged sequentially from bottom to top. A rotating output end of the first rotating driving device is fixed at the center of the rotating disc. A rotating driving end of the second rotating driving device is located at an eccentric position of the rotating disc and connected with a first end of the rotating arm. The second rotating driving device drives the rotating arm to make a circular motion around the first end of the rotating arm as the center of the circular motion. The magnet may adsorb the balls and move the balls inside the sand board.

Inventors: ZHUANG; Zhi (Beijing, CN)

Applicant: Beijing Zhirun Technology Co., Ltd. (Beijing, CN)

Family ID: 92789279

Appl. No.: 19/073430

Filed: March 07, 2025

Foreign Application Priority Data

CN 202420433928.8

Mar. 07, 2024

Publication Classification

Int. Cl.: A63H29/24 (20060101)

U.S. Cl.:

CPC A63H29/24 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This patent application claims the benefit and priority of Chinese Patent Application No. 202420433928.8 filed with the China National Intellectual Property Administration on Mar. 7, 2024, the disclosure of which is incorporated by reference herein in its entirety as part of the present application.

TECHNICAL FIELD

[0002] The present disclosure relates to the technical field of sand painting equipment, in particular to a sand painting machine.

BACKGROUND

[0003] The sand painting is usually created by people sketching the outline on a sand board with wooden sticks. In order to create complex patterns, creators need to be with extremely high skill and operation level.

[0004] With the development of science and technology, sand painting machines have appeared on the market, and the outline is sketched on the sand board by mechanical structures. An existing sand painting machine is disclosed in the Chinese patent entitled as “INTELLIGENT SAND PAINTING MACHINE” with the Granted Patent Publication No. 211843919U. In this patent, a magnet is driven to move by a rotating mechanism and a translation mechanism, and finally balls are adsorbed to draw patterns on the sand board through the movement of the magnet.

[0005] However, the translation mechanism in the existing mechanism is complicated in structure and cumbersome in operation.

[0006] Therefore, there is an urgent need of a novel sand painting machine to solve the above-mentioned problems.

SUMMARY

[0007] The present disclosure aims to provide a sand painting machine so as to solve the above-mentioned technical problem in the prior art. A magnet is driven to move by a first rotating driving device and a second rotating driving device, and finally balls are adsorbed and driven to move inside a sand board by the magnet. The integral structure and working path of the sand painting machine in the present disclosure are simple and convenient.

[0008] In order to achieve the above-mentioned purpose, the present disclosure provides the following technical solutions.

[0009] A sand painting machine is provided in the present disclosure, including a fixed chassis, a rotating disc, a rotating arm and a sand board which are arranged sequentially from bottom to top. The fixed chassis is provided with a first rotating driving device. A rotating output end of the first rotating driving device is fixed at a center of the rotating disc. The first rotating driving device is configured for driving the rotating disc to rotate. The rotating disc is provided with a second rotating driving device. A rotating driving end of the second rotating driving device is located at an eccentric position of the rotating disc. The rotating driving end of the second rotating driving device is connected with a first end of the rotating arm. A second end of the rotating arm is provided with a magnet. The second rotating driving device is configured for driving the rotating arm to make a circular motion around the first end of the rotating arm which acts as a center of the circular motion. Sand and balls are placed in the sand board. The magnet is capable of adsorbing the balls and making the balls move inside the sand board.

[0010] Preferably, the first rotating driving device includes a first driving motor, a first drive wheel and a first driven wheel. The first driving motor is fixed to a lower surface of the fixed chassis. An output shaft of the first driving motor passes through the fixed chassis and is connected with the first drive wheel. The first drive wheel is connected with the first driven wheel through a first

transmission belt. The first driven wheel is located at the centers of the fixed chassis and the rotating disc. A lower surface of the first driven wheel is rotatably connected with an upper surface of the fixed chassis. An upper surface of the first driven wheel is fixed to a lower surface of the rotating disc.

[0011] The second rotating driving device includes a second driving motor, a second drive wheel and a second driven wheel. The second driving motor is fixed to the lower surface of the rotating disc. An output shaft of the second driving motor passes through the eccentric position of the rotating disc. The second drive wheel is fixed on the output shaft of the second driving motor. The second drive wheel is connected with the second driven wheel through a second transmission belt. The second driven wheel is located at the eccentric position of the rotating disc. A lower surface of the second driven wheel is rotatably connected to an upper surface of the rotating disc. An upper surface of the second driven wheel is fixedly connected with the first end of the rotating arm.

[0012] Preferably, the lower surface of the first driven wheel is connected with the upper surface of the fixed chassis through a first universal rotary table; and the lower surface of the second driven wheel is connected with the upper surface of the rotating disc through a second universal rotary table.

[0013] Preferably, the lower surface of the fixed chassis is provided with a plurality of fixed legs.

[0014] Preferably, a first photoelectric displacement limiter is fixed on the fixed chassis, and a lower end of the rotating disc is provided with a first trigger rod; and [0015] a second photoelectric displacement limiter is fixed on the upper surface of the rotating disc, and a lower surface of the second end of the rotating arm is provided with a second trigger rod.

[0016] Preferably, the upper surface of the fixed chassis is provided with a plurality of support rods. Upper ends of all the support rods are configured for supporting a transparent cover. An opening in an upper end of the sand board is covered with the transparent cover.

[0017] Preferably, support sheets are fixed on the support rods in a manner of one-for-one, and the sand board is placed on an upper surface of the support sheet.

[0018] Compared with the prior art, the present disclosure has the following technical effects.

[0019] According to the sand painting machine of the present disclosure, the rotating disc is driven to rotate by the first rotating driving device, and then the rotating arm is driven by the second rotating driving device to make a circular motion around the second end of the rotating arm which acts as the center of the circular motion. The magnet may be driven to move to the predicted position through two nonconcentric circular paths, and finally the magnet is used for adsorbing balls and making the balls draw corresponding images in the sand board. The sand painting machine is simple in overall structure and low in manufacturing cost. The sand painting machine only performs two circular paths in the whole process, and the moving paths are relatively simple and convenient to control.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] To more clearly illustrate the technical solutions in the embodiments of the present disclosure or in the prior art, the following briefly introduces the accompanying drawings to be required and used in the embodiments. Apparently, the accompanying drawings in the following description show merely some embodiments of the present disclosure, and those skilled in the art may still derive other drawings from these accompanying drawings without creative efforts.

[0021] FIG. 1 is a structural schematic diagram of a sand painting machine in the embodiment of the present disclosure.

[0022] FIG. 2 is a structural schematic diagram of a rotating disc in a sand painting machine in the embodiment of the present disclosure.

[0023] FIG. 3 is a structural schematic diagram of a fixed chassis in a sand painting machine in the embodiment of the present disclosure.

[0024] FIG. 4 is a structural schematic diagram of a sand painting machine in another embodiment of the present disclosure.

[0025] Reference numbers in the accompanying drawings: **1**—fixed chassis; **2**—rotating disc; **3**—rotating arm; **4**—sand board; **5**—transparent cover; **6**—fixed leg; **7**—first driving motor; **8**—first drive wheel; **9**—first driven wheel; **10**—first photoelectric displacement limiter; **11**—first trigger rod; **12**—second driving motor; **13**—second drive wheel; **14**—second driven wheel; **15**—second trigger rod; **16**—second photoelectric displacement limiter; **17**—magnet; **18**—support rod; **19**—support sheet; and **20**—rotating arm.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0026] The following clearly and completely describes the technical solutions in the embodiments of the present disclosure with reference to the embodiments of the present disclosure. Apparently, the described embodiments are merely a part rather than all of the embodiments of the present disclosure. Based on the embodiments in the present disclosure, all other embodiments acquired by those skilled in the art under the premise of without creative efforts fall within the scope protected by the present disclosure.

[0027] The present disclosure aims to provide a sand painting machine so as to solve the above-mentioned technical problem in the prior art. A magnet is driven to move by a first rotating driving device and a second rotating driving device, and finally balls are adsorbed and driven to move inside a sand board by the magnet. The integral structure and working path of the sand painting machine in the present disclosure are simple and convenient.

[0028] To make the foregoing objective, features and advantages of the present disclosure clearer and more comprehensible, the following further describes the present disclosure in detail with reference to the accompanying drawings and specific embodiments.

[0029] As shown in FIG. 1 to FIG. 3, this embodiment provides a sand painting machine. The sand painting machine includes a fixed chassis **1**, a rotating disc **2**, a rotating arm **3** and a sand board **4** which are arranged sequentially from bottom to top. The fixed chassis **1**, the rotating disc **2** and the sand board **4** are all of circular-disc structures. The fixed chassis **1** is provided with a first rotating driving device. A rotating output end of the first rotating driving device is fixed at the center of the rotating disc **2**. The first rotating driving device is configured for driving the rotating disc **2** to rotate, so that each part on the rotating disc **2** may move along with the rotation of the rotating disc **2**. The rotating disc **2** is provided with a second rotating driving device. A rotating driving end of the second rotating driving device is located at an eccentric position of the rotating disc **2**. A rotating driving end of the second rotating driving device is connected with a first end of the rotating arm **3**. A second end of the rotating arm **3** is provided with a magnet **17**. As shown in FIG. 2, the first end of the rotating arm **3** is larger in size than that of the second end thereof. The second rotating driving device is configured for driving the rotating arm **3** to make a circular motion around the first end of the rotating arm **3** which acts as the center of the circular motion. Sand and balls are placed in the sand board **4**. The ball is an iron ball. The magnet **17** may adsorb the balls to move inside the sand board **4**.

[0030] As shown in FIG. 4, in another embodiment, the approximately redundant portion of the rotating disk **2** (e. g., portions other than portions required to fix the first rotating driving device and the second rotating driving device, without including portions required for normal transmission of force) can be removed to produce a rotating arm **20**. One end of the rotating arm **20** may be equivalent to the center area portion of the rotating disk **2**, and the other end of the rotating arm **20** may be equivalent to the eccentric region portion of the rotating disk **2** (where the second rotating driving device is provided).

[0031] In actual use, the first rotating driving device is turned on. The rotating disc **2**, the second rotating driving device on the rotating disc **2**, the rotating arm **3** and other structures are driven by

the first rotating driving device to make a circular motion around the center of the rotating disc 2 which acts as the center of the circular motion. During this period, the second rotating driving device is turned on. The rotating arm 3 and the magnet 17 are driven by the second rotating driving device to make a circular motion around the first end of the rotating arm 3 which acts as the center of the circular motion. Through two circumferential paths of the first rotating driving device and the second rotating driving device, the magnet 17 may be driven to move according to the predicted path, so that the balls are adsorbed to draw sand painting patterns on the sand board 4.

[0032] In the embodiment, as shown in FIG. 3, the first rotating driving device includes a first driving motor 7, a first drive wheel 8 and a first driven wheel 9, wherein a shell of the first driving motor 7 is fixed to a lower surface of the fixed chassis 1 through screws. An output shaft of the first driving motor 7 passes through a corresponding through hole in the fixed chassis 1 and then is connected with the first drive wheel 8. The first drive wheel 8 is connected in a transmission manner with the first driven wheel 9 through a first transmission belt. The diameter of the first drive wheel 8 is smaller than that of the first driven wheel 9. The fixed chassis 1 is coaxially arranged with the rotating disc 2. The first driven wheel 9 is located at the centers of the fixed chassis 1 and the rotating disc 2. A lower surface of the first driven wheel 9 is rotatably connected with an upper surface of the fixed chassis 1. An upper surface of the first driven wheel 9 is fixed to a lower surface of the rotating disc 2 through multiple screws.

[0033] When there is a need to turn on the first rotating driving device, it is only needed to turn on the first driving motor 7. The first driving motor 7 drives the first drive wheel 8 to rotate. The first drive wheel 8 drives the first driven wheel 9 to rotate through the first transmission belt, and the first driven wheel 9 finally drives the rotating disc 2 to rotate since the first driven wheel 9 is fixedly connected with the center of the rotating disc 2.

[0034] Similar to the first rotating driving device, as shown in FIG. 2, the second rotating driving device includes a second driving motor 12, a second drive wheel 13 and a second driven wheel 14. A shell of the second driving motor 12 is fixed to the lower surface of the rotating disc 2 through screws, and an output shaft of the second driving motor 12 passes through the eccentric position of the rotating disc 2. The second drive wheel 13 is fixed to an output shaft of the second driving motor 12, that is, the second driving motor 12 may drive the second drive wheel 13 to rotate. The second drive wheel 13 is connected in a transmission manner with the second driven wheel 14 through a second transmission belt. The diameter of the second driven wheel 14 is larger than that of the second drive wheel 13. The second driven wheel 14 is located at the eccentric position of the rotating disc 2. A lower surface of the second driven wheel 14 is rotatably connected to an upper surface of the rotating disc 2, and an upper surface of the second driven wheel 14 is fixedly connected with the first end of the rotating arm 3 through multiple screws.

[0035] Similarly, when there is a need to turn on the second rotating driving device, it is only needed to turn on the second driving motor 12. The second driving motor 12 drives the second drive wheel 13 to rotate. The second drive wheel 13 drives the second driven wheel 14 to rotate through the second transmission belt, and the second driven wheel 14 finally drives the magnet 17 on the second end of the rotating arm 3 to make a circular motion around the first end of the rotating arm 3 which acts as the center of the circular motion since the second driven wheel 14 is fixedly connected with the first end of the rotating arm 3.

[0036] In the further embodiment, a controller may also be additionally arranged. The first driving motor 7 and the second driving motor 12 are both electrically connected with the controller for controlling the operation of the first driving motor 7 and the second driving motor 12. The controller includes, but not limited to an existing single-chip microcomputer controller or a PLC (Programmable Logic Controller).

[0037] In the embodiment, in order to realize the rotational connection between the first driven wheel 9 and the fixed chassis 1, the lower surface of the first driven wheel 9 is connected with the upper surface of the fixed chassis 1 through a first universal rotary table.

[0038] Similarly, the lower surface of the second driven wheel **14** is connected with the upper surface of the rotating disc **2** through a second universal rotary table.

[0039] In the embodiment, both the first universal rotary table and the second universal rotary table may be universal rotary tables (or called as universal discs) generally installed the lower portion of a rotating chair in the prior art, and the universal rotary table includes an upper disc body and a lower disc body. Multiple bearing balls are arranged between the upper disc body and the lower disc body so as to realize the relative rotation of the upper disc body and the lower disc body.

[0040] In the embodiment, the upper disc body of the first universal rotary table is fixed with the lower surface of the first driven wheel **9** through screws, and the lower disc body of the first universal rotary table is fixed with the fixed chassis **1** through screws. The upper disc body of the second universal rotary table is fixed with the lower surface of the second driven wheel **14** through screws, and the lower disc body of the second universal rotary table is fixed with the rotating disc **2** through screws.

[0041] In the embodiment, the lower surface of the fixed chassis **1** is provided with multiple fixed legs **6**, specifically four fixed legs, and the four fixed legs **6** are equally distributed on the lower surface of the fixed chassis **1** along the circumferential direction. The fixed chassis **1** and all parts arranged on the fixed chassis **1** may be supported by the fixed legs **6**.

[0042] In the embodiment, a first photoelectric displacement limiter **10** is fixed on the fixed chassis **1**. The first photoelectric displacement limiter **10** may be a photoelectric displacement limiter **10** with an existing U-shaped structure. A lower end of the rotating disc **2** is provided with a first trigger rod **11**. The first trigger rod **11** is vertical to the rotating disc **2**. When the first trigger rod **11** passes through the first photoelectric displacement limiter **10**, the first trigger rod **11** may pass through the middle of the first photoelectric displacement limiter **10**, thereby triggering the first photoelectric displacement limiter **10**. At this time, the position of the rotating disc **2** is defined as an initial position thereof.

[0043] Similarly, a second photoelectric displacement limiter **16** is fixed on the upper surface of the rotating disc **2**, and a second trigger rod **15** is arranged on a lower surface of the second end of the rotating arm **3**. The second trigger rod **15** is vertical to a lower surface of the rotating arm **3**. When the second trigger rod **15** passes through the second photoelectric displacement limiter **16**, the second trigger rod **15** may pass through the middle of the second photoelectric displacement limiter **16**, thereby triggering the second photoelectric displacement limiter **16**. At this time, the position of the rotating arm **3** is defined as an initial position thereof.

[0044] In the embodiment, the upper surface of the fixed chassis **1** is provided with several support rods **18**, specifically four support rods **18**, and the support rods **18** are equally distributed on the fixed chassis **1** along the circumferential direction. Upper ends of all the support rods **18** may be used for support a transparent cover **5**. The materials of the transparent cover **5** include, but not limited to glass or transparent plastics. An opening in an upper end of the sand board **4** is covered with the transparent cover **5**. The transparent cover **5** can facilitate people to directly observe internal situations of the sand board **4** through the transparent cover **5**, and may also avoid the sand and balls inside the sand board **4** from running out.

[0045] In the embodiment, support sheets **19** are fixed on the support rods **18** in a manner of one-for-one. Upper and lower sides of each support sheet **19** are respectively in threaded connection with one nut. The height of the support **19** may be adjusted through the two nuts. The sand board **4** is placed on an upper surface of the support sheet **19**, so that the sand board **4** may be supported. Further, two threaded holes may be formed in the support sheet **19** for fixing the sand board **4** through screws, thereby achieving a technical effect of further stabilizing the sand board **4**.

[0046] Specific examples are used in the present disclosure for illustration of the principles and embodiments of the present disclosure. The description of the above-mentioned embodiments is only used to help understand the method and its core ideas of the present disclosure. In addition, those skilled in the art can make various modifications in terms of specific embodiments and scope

of application in accordance with the ideas of the present disclosure. In summary, the contents of this specification should not be understood as a limitation to the present disclosure.

Claims

1. A sand painting machine, comprising a fixed chassis, a rotating arm, a rotating arm and a sand board which are arranged sequentially from bottom to top, wherein the fixed chassis is provided with a first rotating driving device, a rotating output end of the first rotating driving device is provided at a center of the fixed chassis and fixed with one end of the rotating arm, the first rotating driving device is configured for driving the rotating arm to rotate, the rotating arm is provided with a second rotating driving device, a rotating driving end of the second rotating driving device is located at an other end of the rotating arm, the rotating driving end of the second rotating driving device is connected with a first end of the rotating arm, a second end of the rotating arm is provided with a magnet, the second rotating driving device is configured for driving the rotating arm to make a circular motion around the first end of the rotating arm which acts as a center of the circular motion, sand and balls are placed in the sand board, and the magnet is capable of adsorbing the balls and making the balls move inside the sand board.

2. The sand painting machine according to claim 1, wherein the first rotating driving device comprises a first driving motor, a first drive wheel and a first driven wheel, the first driving motor is fixed to a lower surface of the fixed chassis, an output shaft of the first driving motor passes through the fixed chassis and is connected with the first drive wheel, the first drive wheel is connected with the first driven wheel through a first transmission belt, the first driven wheel is located at the centers of the fixed chassis, a lower surface of the first driven wheel is rotatably connected with an upper surface of the fixed chassis, and an upper surface of the first driven wheel is fixed to a lower surface of the rotating arm; and the second rotating driving device comprises a second driving motor, a second drive wheel and a second driven wheel, the second driving motor is fixed to the lower surface of the rotating arm, an output shaft of the second driving motor passes through the other end of the rotating arm, the second drive wheel is fixed on the output shaft of the second driving motor, the second drive wheel is connected with the second driven wheel through a second transmission belt, the second driven wheel is located at the other end of the rotating arm, a lower surface of the second driven wheel is rotatably connected to an upper surface of the rotating arm, and an upper surface of the second driven wheel is fixedly connected with the first end of the rotating arm.

3. The sand painting machine according to claim 2, wherein the lower surface of the first driven wheel is connected with the upper surface of the fixed chassis through a first universal rotary table; and the lower surface of the second driven wheel is connected with the upper surface of the rotating arm through a second universal rotary table.

4. The sand painting machine according to claim 1, wherein the lower surface of the fixed chassis is provided with a plurality of fixed legs.

5. The sand painting machine according to claim 1, wherein a first photoelectric displacement limiter is fixed on the fixed chassis, and a lower end of the rotating arm is provided with a first trigger rod; and a second photoelectric displacement limiter is fixed on the upper surface of the rotating arm, and a lower surface of the second end of the rotating arm is provided with a second trigger rod.

6. The sand painting machine according to claim 1, wherein the upper surface of the fixed chassis is provided with a plurality of support rods, upper ends of all the support rods are configured for supporting a transparent cover, and an opening in an upper end of the sand board is covered with the transparent cover.

7. The sand painting machine according to claim 6, wherein support sheets are fixed on the support rods in a manner of one-for-one, and the sand board is placed on an upper surface of the support

sheet.

8. A sand painting machine, comprising a fixed chassis, a rotating disc, a rotating arm and a sand board which are arranged sequentially from bottom to top, wherein the fixed chassis is provided with a first rotating driving device, a rotating output end of the first rotating driving device is fixed at a center of the rotating disc, the first rotating driving device is configured for driving the rotating disc to rotate, the rotating disc is provided with a second rotating driving device, a rotating driving end of the second rotating driving device is located at an eccentric position of the rotating disc, the rotating driving end of the second rotating driving device is connected with a first end of the rotating arm, a second end of the rotating arm is provided with a magnet, the second rotating driving device is configured for driving the rotating arm to make a circular motion around the first end of the rotating arm which acts as a center of the circular motion, sand and balls are placed in the sand board, and the magnet is capable of adsorbing the balls and making the balls move inside the sand board.
